

Alert Fatigue

Is a Misdiagnosis

Kim Mahan, CISSP · Founding Apprentice & CEO,
MAXX Potential
@kimmahan

"20+ years in technology and I still got hacked."



Why I Care About Cognitive Decline



Alzheimer's took three people I loved. I want to see the two new loves of my life grow & thrive.

I. Was. Hacked.

2022. CISSP. 20+ years IT experience. Running a company built on developing human potential.

- Pandemic information overload
- Social media doom-scroll cycles
- Life transitions + empty nester
- Then ChatGPT dropped ...
- **Couldn't focus. Barely answering emails.**

*"It happened to me.
Could it be happening on your team?"*



The Human OSI Stack

Seven layers. Every one of them has an attack surface.

LAYER	RUNS ON	WHAT IT HANDLES	EXPLOITED BY
7 APPLICATION	<i>Dopamine</i>	Motivation, reward, goal-seeking	HIJACKED Infinite scroll, gamified feeds
6 PRESENTATION	<i>Serotonin</i>	Mood — how your world gets rendered	HIJACKED Comparison & outrage feeds
5 SESSION	<i>Oxytocin</i>	Trust & connection — human to human	THE LAYER WE QUIETLY DEPRECATED STARVED Everything screen-mediated
4 TRANSPORT	<i>Cortisol</i>	Load management under stress	FLOODED Alert storms, breaking news
3 NETWORK	<i>GABA</i>	Calm, inhibition, stable routing	DEGRADED No downtime, ever
2 DATA LINK	<i>Norepinephrine</i>	Attention, alertness, vigilance	SPOOFED Notifications & interrupts
1 PHYSICAL	<i>Endorphins</i>	The hardware — body, movement, sleep	STARVED Ten hours in a chair

"You can't patch a stack you don't know you're running."

A conceptual framework, not a clinical model — these systems overlap and interact constantly.

Alert fatigue is
a symptom, not the diagnosis.

*We Created an Entire Industry
That Trains Humans to Be Pattern
Matching Machines, and AI is
Coming For That Job.*

These Aren't HR Stats. They're Security Risk Metrics.

71%

of SOC analysts
report burnout
(Tines Voice of the SOC)

18–24

months avg analyst
tenure

4,484

avg alerts per
day per team
(Vectra AI, 2023)

1 in 3 hours

time spent on
non-genuine threats
(IBM/Morning Consult Global SOC
study, 2023)

64%

considering leaving
their organization
(Tines Voice of the SOC)

10yrs

of SIEM tuning &
new platforms —
volume still climbs

We've been treating the symptom. The disease is something else.

ABOUT MAXX

Building the Workforce the Industry Needs

Founded 2012 to address the domestic tech talent shortage

IT Consulting + Talent Development via Apprenticeship &
Work-based Learning

Recognized by Fast Company & Southern Living as one of the
"Innovations Changing the South"

"Built by tech leaders for tech leaders."

"Learning By Doing" is in the fabric of our DNA"



Alerts? Yes, We've Got Them!

We run a 24/7 shared services team including security operations, IT support, automation. Entry-level talent, apprentice model, real client work from day one.

The Problem We Saw

A workforce pipeline people trained to consume knowledge, rather than create it.
Certifications without craft.
Degrees without depth.

What We Built Instead

Learn by doing. Skills-based progression. Raises tied to demonstrated creation skills and value-generation; not certs, not tickets closed. Perform harder tasks on the fly.

What We've Discovered

The cognitive atrophy problem shows up in our team too. And fixing it is changing how we run our operation.

Feel Your Own Decline

1

Take out your phone. Open the 'guilty pleasure' app you check most.
Scroll for exactly 60 seconds.

2

Put the phone face down.

3

Tell the person next to you the most important thing you just saw.
Something specific – a name, a number, a claim.

*"That's not a memory problem. That's consumption mode.
Your brain processed it as low-stakes pattern-matching — not information worth encoding."*

The Research Is Converging

47 sec

Avg time before a knowledge worker self-interrupts
(Gloria Mark, UC Irvine, 2023)

 **12.7%**

Error risk per interruption, triples after 6
(Westbrook et al., *Arch Intern Med*, 2010)

**23
minutes**

Average time to regain focus after an interruption
(Gloria Mark, UC Irvine, 2023)

worse

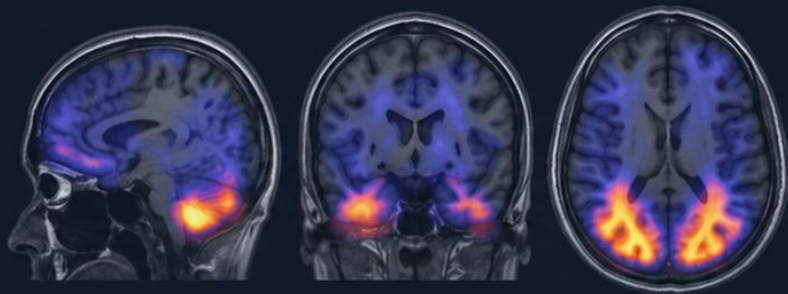
Heavy multitaskers score measurably lower on attention control tests (Ophir/Nass/Wagner, *PNAS*, 2009)

These are the conditions SOC analysts work in every single day.

Same Analyst. Different Work. Different Brain.

CONSUMPTION MODE

Triage / Feed Scroll / Alert Monitoring



PREDICTED ACTIVITY

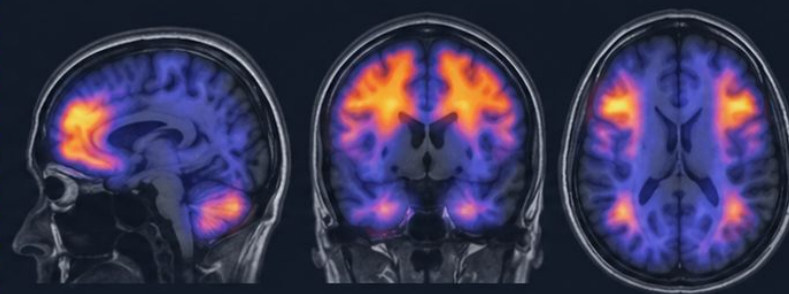
Heightened visual cortex, posterior cingulate,
and parietal regions.

Lower prefrontal engagement.

*Passive consumption. Narrow focus.
Minimal executive load.*

CREATION MODE

Threat Hunting / Building / Deep Analysis



PREDICTED ACTIVITY

Increased prefrontal cortex, anterior cingulate,
and **insula** activation.

Stronger networks for reasoning and decision-making.

*Active creation. Deep focus.
Higher cognitive integration and control.*

Source: Meta TRIBE v2 — predicted average neural response across 700+ subjects (not individual scan)

The Patch I Applied

Before: what it looked like

- Reading same email 3× without retaining it
- Switching tabs every 90 seconds
- Felt productive — was just busy
- Couldn't hold a thought through a meeting

What had to change

- Touched grass — deliberately offline, moving, outside
- Dance: hits every layer of the human stack
- Information diet — I know when I'm eating junk food
- Sharing this publicly. Creation is the medicine.



My New & Improved Information Diet

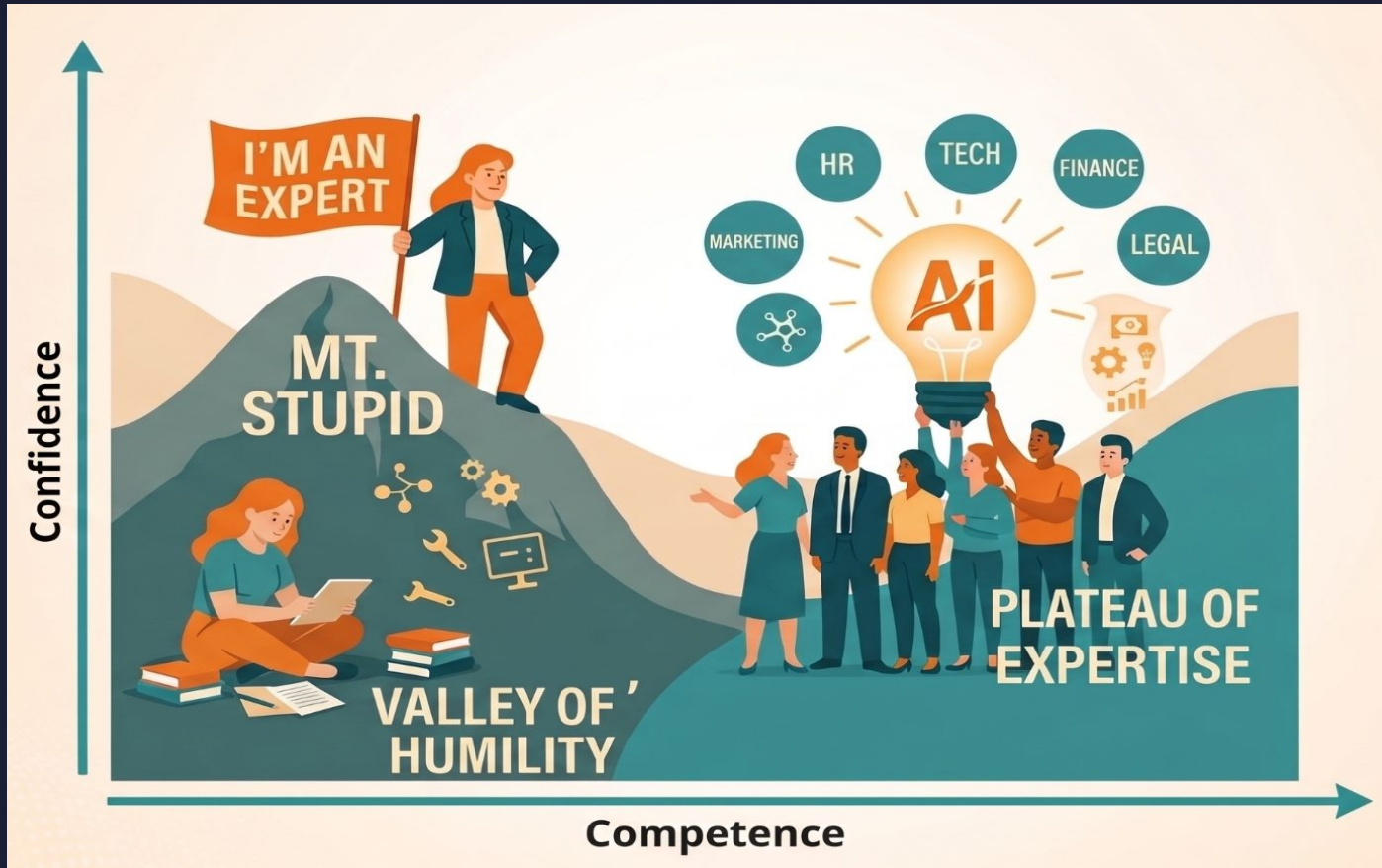
Calories = Minutes; Consume and Create in Equal Measure

Information Type	Cognitive Effect
Social scrolling	Sugar
Breaking news	Caffeine
Vendor webinars	Processed food
Books/Writing	Protein
Deep conversations	Whole foods
Building something	Strength training

"If your team's information diet looks like their lunch plate, would you be proud?"



AI-Induced Dunning-Kruger Effect



AI Is Creating a Massive Dunning-Kruger Effect

The Danger

- Outsource thinking to AI outputs
- Can't explain what AI generated
- Flawed outputs accepted uncritically
- High confidence, low actual depth
- Mount Stupid at scale

The Opportunity

- AI frees cognitive space
- Only if you maintained depth
- Creation-mode teams thrive
- Triage-only teams have nothing left
- Plateau of Expertise — earned

Epistemic passivity: the technical term for what alert triage trains.

Feel the Difference

Turn to the person next to you.

One person asks:

"When was the last time you were genuinely absorbed in a hard problem at work — where you lost track of time?"

90 seconds each. Then switch. No phones. Eye contact.

Notice what your nervous system feels like right now
compared to the scrolling exercise.

AI Is Automating the Work That Was Making Us Numb.

AI is automating:

- Alert triage
- Correlation & deduplication
- Ticket routing
- Threat feed summarization
- Pattern matching at scale

What remains:

- Novel threat detection
- Reading attacker intent in ambiguous data
- Creative defense architecture
- Building custom detections
- Adversarial creative thinking

AI is removing what atrophied your team . . . and exposing what atrophy destroyed.

The Consumption Audit

Our team found 75–80% of our analysts time is reactive consumption.
Leaders (including me) are shocked by the number. That number is a security risk metric.

1

What % of your team's day is someone else's agenda?

(alerts, tickets, meetings, email)

2

What did your team build or create last week?

(a detection, a process, anything new that didn't exist before)

3

When did your analysts last feel absorbed in a hard problem?

(if the answer doesn't come fast — that's data)

Frame this to leadership as a security risk audit — not a wellness initiative.

Downtime Is Maker Time, Not More Consumption

The Intelligent Automation Board

- Kanban board of real build projects and harder analytical challenges
- Shared services works on maker projects during low-volume time
- No triage during maker blocks; interruptions are the enemy
- Projects rotate to build team breadth

Skills-Based Level-Up

- Raises tied to demonstrated creative work and skills, not tickets closed
- Apprentices earn levels by performing harder tasks on the fly
- Novel detection, custom builds, original analysis = the metrics
- What you practice toward is what you become

We caught what the automated system missed – not at peak volume, but on weekends when analysts had cognitive space.

Creation-Mode Ideas and Practices for Security Teams

Threat hunting blocks

Calendar-blocked, non-interruptible. No alert triage during them. Treat them as seriously as incident response.

Build, don't only consume

Custom detections over vendor rules. Original runbooks. Novel analysis. Build things that don't exist yet.

Maker rotation

One analyst per cycle owns a build project — zero triage duty. Rotate who's in maker mode. Interrupt the consumption-only pattern.

Measure creation output

Track what gets built, not just processed. Custom detections, hunt findings, original analysis — these are metrics too.

The Person Who Needs to Change First Might Be You.

What makes it worse:

- ↓ More metrics, more pressure
- ↓ Public performance dashboards
- ↓ Measuring only throughput
- ↓ Rewarding reactive speed over depth
- ↓ Leaders who can't focus either

Operational empathy:

- ↑ Protect cognitive space, don't just measure throughput
- ↑ Shield maker time from interruption
- ↑ Name the problem to your team
- ↑ Run the consumption audit on yourself first
- ↑ Model creation-mode work visibly

"The leader who most needs to change is often the one who built their career on consumption-mode efficiency."

Run Your Own Audit Right Now

1 What % of your last workday was someone else's agenda?

(alerts, tickets, meetings, email. Be honest)

2 What did you build or create that didn't exist before you touched it? *(detection, doc, process, anything new)*

3 When did you last feel absorbed in a hard problem?

(if the answer doesn't come fast, that's data)

Don't judge the number. Just see it. That's your baseline.



Kim Mahan
kim@maxxpotential.com
@kimmahan
MAXX Potential